

ISHAAN RANJAN

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EDUCATION

Georgia Institute of Technology

Expected Dec 2028

B.S. Computer Science | Incoming Fall 2026

- Accelerated 2.5-year plan supported by AP/transfer credit; planned coursework focus: algorithms, artificial intelligence, systems, and theory.

BASIS Peoria

May 2026

High School Diploma | GPA: 3.99 UW / 4.87 W | SAT: 1540 | ACT: 35

- Completed 15 AP exams (14 scores of 5, one score of 4); received the \$5,000 BASIS Educational Ventures Scholarship.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Swift, SQL, R, HTML/CSS

AI/ML: PyTorch, Hugging Face Transformers, PEFT/LoRA, scikit-learn, Whisper, RAG, embeddings, LLM evaluation

Systems & Product: FastAPI, React/Next.js, Electron, SwiftUI, Supabase/PostgreSQL, Docker, GitHub Actions, Arduino

PROFESSIONAL EXPERIENCE

Luxen AI — Co-Founder & Engineering Lead | luxenai.org

Dec 2023–Present

- Co-founded an applied-AI product studio and led product/technical execution from user discovery through architecture, evaluation, deployment, demos, and partner communication; conducted 100+ user interviews to prioritize real workflows over feature speculation.
- Projects reached 5,000+ active users and 1.5M+ media views; secured \$48K+ in grants, startup support, and cloud credits and earned selection as a two-time Z Fellows finalist and 1517 Medici participant.
- Built developer tools, agent infrastructure, on-device AI prototypes, research pipelines, and production web applications while keeping research prototypes and unvalidated claims explicitly labeled.

Precision Aging Network Lab — Research Intern

Summer 2025

- Analyzed Horvath, AltumAge, and DunedinPACE biological-aging measures in R/Python across vitamin D, omega-3, and exercise interventions; performed exploratory statistics and visualization while documenting null findings and confounding constraints.
- Presented methodology and results to 800+ attendees through the KEYS Research Internship program.

ENGINEERING PROJECTS

Regimen (Luxen AI) — Small-Model Agent Training & Evaluation | GitHub

2025–Present

Research prototype and open-source evaluation harness | Python, PyTorch, Transformers, PEFT/LoRA

- Co-designed a 200-task controlled tool-use study: fine-tuned Qwen2.5-3B with a custom harness reached 94.5% task success versus 95.5% for Qwen2.5-72B, lowering cost per successful task by 84.8%.
- Audited the later A100 integration and found deterministic paths mislabeled as SLM inference, item-type schema leakage, and synthetic-data generalization failure; repaired validation and documented the invalidated claims.
- Established retrieval baselines and failure tests; BM25 improved recall@5 from 5/10 to 7/10 with 85.8% less context, creating an interpretable benchmark for learned methods.

YConstruction — Voice-First, Offline Construction Reporting | GitHub | Demo

2026

- Built a SwiftUI field app running Whisper, Gemma 3n E2B, and Qwen3 embeddings fully on device; converts voice/photos into IFC-constrained defects, retrieves local project context, and emits BCF for architect review.
- Implemented a disk-first offline queue, conflict-aware Supabase sync, and Blender/Bonsai review loop; presented the working system live to YC investors at the YC × Google DeepMind hackathon.

Arbor — Local-First Multi-Agent Workspace | GitHub

2025–2026

- Co-built an Electron/TypeScript workspace for master/subagent sessions, shared context, runtime model switching, and human review; kept execution local without requiring a hosted backend.
- Explored orchestration, context transfer, parallel work, and developer control through a functional prototype rather than claiming production-scale deployment.

Debatica — Debate Research Platform

2024–2025

- Built and operated a production web application used by 2,000+ debaters, with crawling, deduplication, source organization, and search workflows for evidence research.
- Translated tournament research habits into product requirements and iterated the ingestion and retrieval workflow around active user feedback.

ADDITIONAL ENGINEERING PROJECTS**A.C.R.E. / GreenhouseAgent — Autonomous Greenhouse Prototype** 2025

- Integrated Arduino sensors and pump control with FastAPI, OpenAI-assisted planning, a local Random Forest, Supabase persistence, and a monitoring dashboard; built as an MIT Blueprint hackathon prototype.
- Implemented the full sensor-to-decision loop and surfaced environmental readings and actuator state in a live dashboard for rapid operator review.

RESEARCH & PUBLICATIONS**PRISM — Policy Evaluation & Forecasting System** | GitHub | Dashboard | Scholar 2025–Present*Research system and preprint | Causal inference, NLP, forecasting, public dashboard*

- Constructed a 51-state, 2000–2022 alcohol-policy panel and event-study difference-in-differences pipeline with 15+ automated figures; documented parallel-trends and causal-identification limits.
- Built a mechanism classifier with macro-F1 0.962 on a 250-row audited benchmark and a held-out 2020–2023 Random Forest forecast with RMSE 0.863 and R^2 0.454.
- Published an interactive dashboard exposing data, methods, figures, and model outputs; manuscript remains a preprint rather than a peer-reviewed publication.

Nemotron Elastic Reproduction — Model Compression Audit 2026*Work in progress | NVIDIA Nemotron Nano 2 VL*

- Reproduced the official 12B parent and supported 9B/6B elastic subnetworks on an A100, then documented width-slicing behavior and budget limits; withheld unsupported size/quality claims pending complete evaluation.
- Separated confirmed reproduction results from proposed extensions and preserved commands, configurations, and failure notes for later benchmarking.

S.A.B.E.R. — Sensor-Controlled Bioreactor | Scholar 2024–2025*Environmental engineering prototype and preprint*

- Designed a bioreactor prototype with sensor-driven pump control for nutrient recovery; earned an Arizona Science and Engineering Fair Grand Award and documented the system in a public preprint.
- Integrated sensing, control logic, and experimental logging into a repeatable benchtop workflow and reported prototype constraints alongside results.

Criminal Tribes Act — Historical Research Paper | Scholar 2024

- Produced an archival analysis of colonial legal classification in India; ranked first in its SSRN download category and qualified for National History Day nationals.

PharmaFlow — Drug-Shortage Risk Modeling 2026*Senior capstone prototype*

- Developed a shortage-risk classifier that reached 82% accuracy on a constructed dataset designed to mirror CMS fields; labeled results as prototype evidence, not clinical or operational validation.
- Built the data-processing and model-evaluation workflow to compare supply-risk signals while keeping synthetic-data limitations visible in the final presentation.

LEADERSHIP & COMMUNITY**BASIS Peoria Speech & Debate — Captain** 2023–2026

- Led a 110+ member team; Arizona State Champion, NSDA Big Questions Top 5, and national qualifier; coached younger competitors in research, argument design, and speaking.

Research Club — President | Science Olympiad — Engineering Lead 2023–2026

- Mentored 30+ students through 10+ research papers and led engineering builds; earned 1st place in Wind Power.

AI Literacy Workshops — Organizer & Instructor 2024–2026

- Designed and delivered practical AI-literacy sessions reaching 150+ students and community members, emphasizing evaluation, limitations, privacy, and responsible use.

SELECTED DISTINCTIONS

- Debate: Arizona State Champion; NSDA Big Questions Top 5 nationally; NSDA Nationals qualifier; Captain of a 110+ member team.
- Research & Entrepreneurship: BASIS Educational Ventures Scholarship (\$5,000); two-time Z Fellows Finalist; 1517 Medici; YC Startup School; AZSEF Grand Award; National History Day Nationals Qualifier.